Reinforcement Learning

from one delivery van to the Bellman equation

Carlos Cotrini, ETH Zurich



Rembrandt, *Faust* (etching, c. 1652)

Faust, alone in his study, is translating the opening of the Gospel of John, from the Greek:

Ἐν ἀρχῇ ἦν ὁ λόγος

"In the beginning was the *logos*."

One Greek word, *logos*, with no single match: it can mean Word, and also Thought, Reason, Power, the Deed.

So he writes a rendering, strikes it out, and tries again:



J. K. Stieler, *Goethe* (1828)

'Tis written: "In the Beginning was the Word."
Here am I balked: who, now, can help afford?
The Word? Impossible so high to rate it;
And otherwise must I translate it,
If by the Spirit I am truly taught.
Then thus: "In the Beginning was the Thought."
This first line let me weigh completely,
Lest my impatient pen proceed too fleetly.
Is it the Thought which works, creates, indeed?
"In the Beginning was the Power," I read.
Yet, as I write, a warning is suggested,
That I the sense may not have fairly tested.
The Spirit aids me: now I see the light!
"In the Beginning was the Act," I write.

Goethe, *Faust I* (1808), trans. Bayard Taylor

the Word *(das Wort)*	→	week one: observing, LLMs
the Thought *(der Sinn)*	→	week two: thinking
the Power *(die Kraft)*	→	the false summit: raw compute
the Act *(die Tat)*	→	week three: acting, today



MDP maze



bandit arcade



Bellman's house



SARSA maze



darts in the dark

The plan

How do we pick the best call?

the one with the largest long-run reward

How do we know that reward?

the Bellman equation gives $Q^*(s, a)$

And when the odds are hidden?

learn it from experience: SARSA

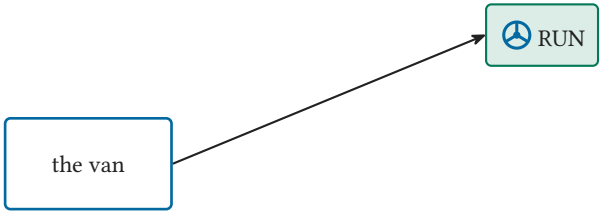
Motivation

One van, one call a week



every week: exactly one call

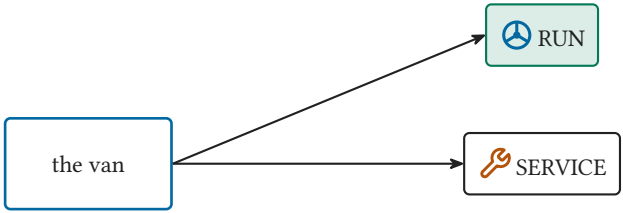
One van, one call a week



collect the week's profit
risk: breakdown, wear

every week: exactly one call

One van, one call a week

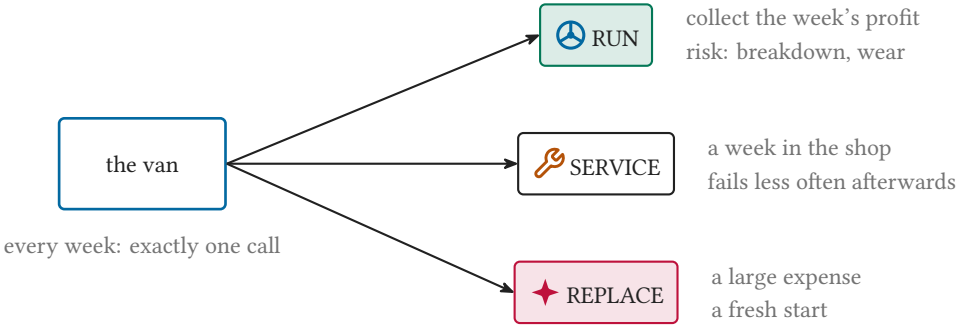


collect the week's profit
risk: breakdown, wear

a week in the shop
fails less often afterwards

every week: exactly one call

One van, one call a week



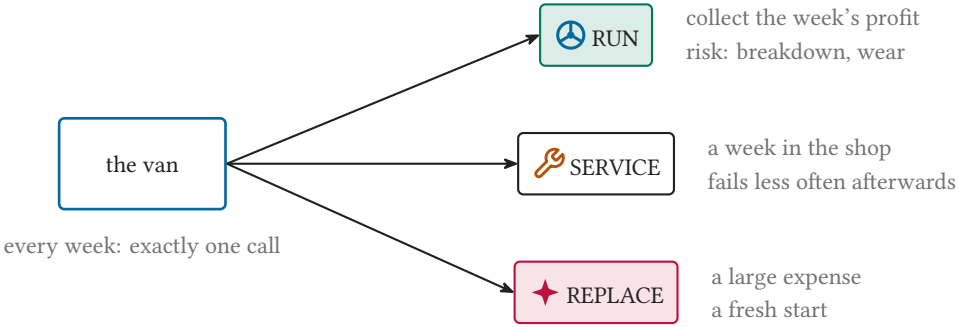
every week: exactly one call

collect the week's profit
risk: breakdown, wear

a week in the shop
fails less often afterwards

a large expense
a fresh start

One van, one call a week



short-term cost against long-term payoff, chance at every step

Markov decision processes



MDP maze



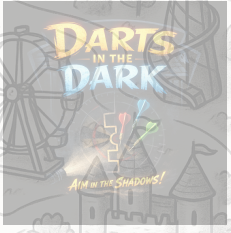
bandit arcade



Bellman's house



SARSA maze



darts in the dark

you are here

What are we deciding, and against what?

one call a week (run, service, replace) in a world that answers at random: a Markov decision process

The model: four ingredients

S: states

four wear levels

The model: four ingredients

S: states

four wear levels

A: actions

three calls

The model: four ingredients

S: states	A: actions	T: dynamics
four wear levels	three calls	how the world reacts

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four wear levels	three calls	how the world reacts	how much the future counts

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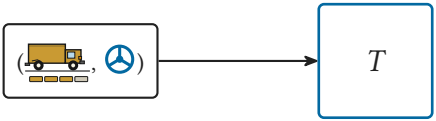
$$(s', r) = T(s, a)$$

The model: four ingredients

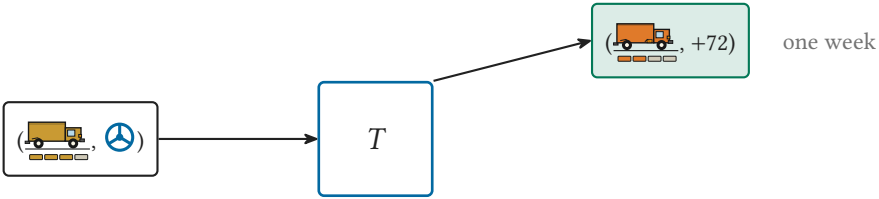
S: states	A: actions	T: dynamics	$\gamma = 0.9$
four wear levels	three calls	how the world reacts	how much the future counts

$$(\underbrace{s'}_{\text{the state next week}}, \underbrace{r}_{\text{the reward}}) = T(s, a)$$

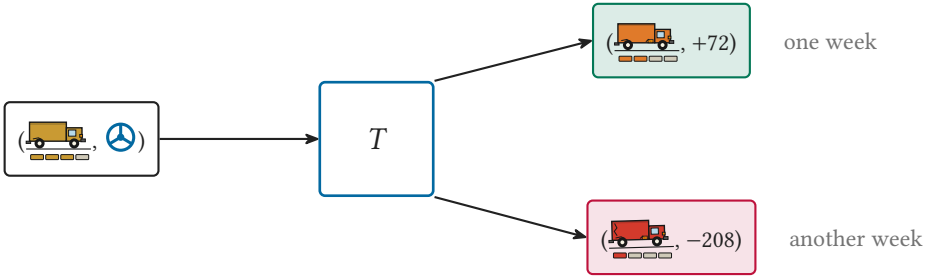
Same pair in, different output



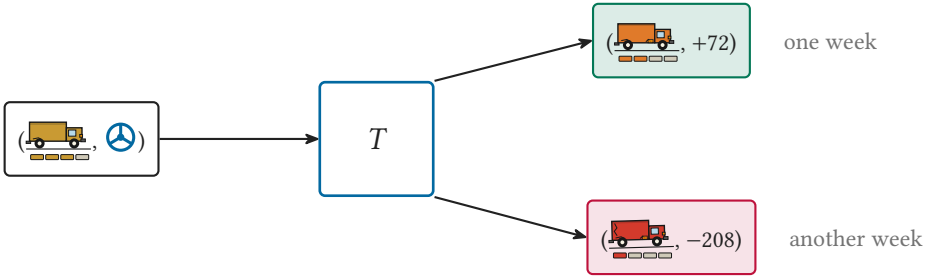
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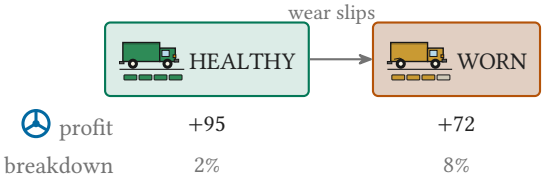
same input, different output: the dynamics T is random

The fleet sheet

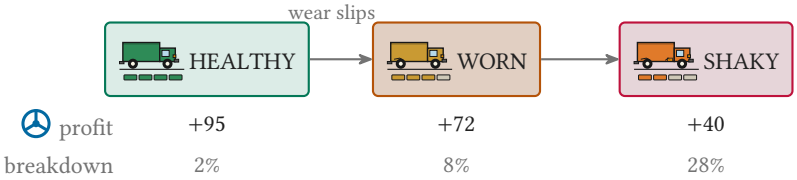


profit	+95
breakdown	2%

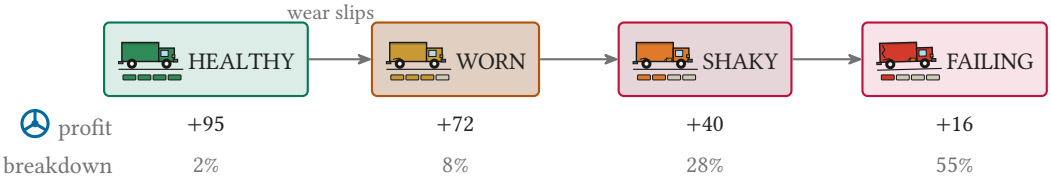
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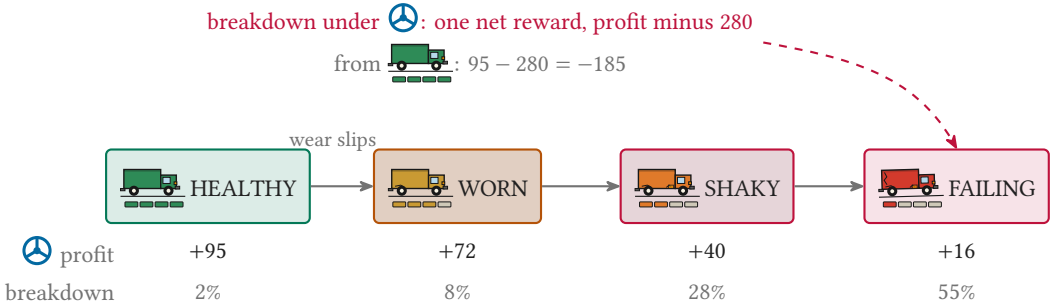
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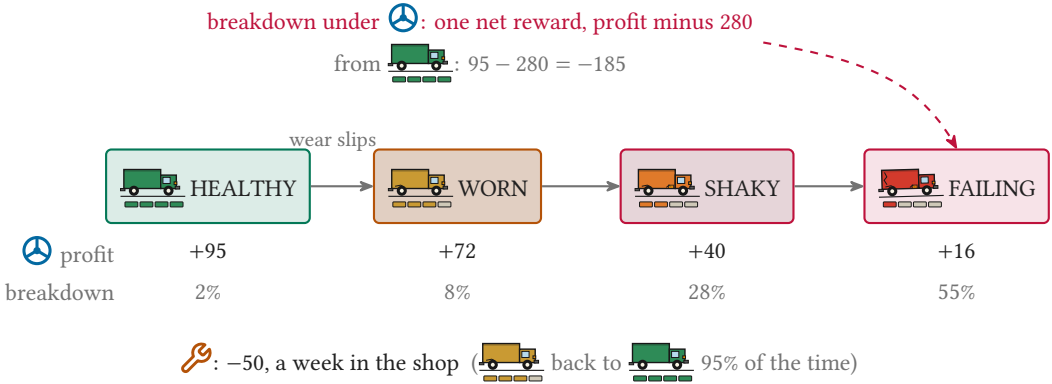
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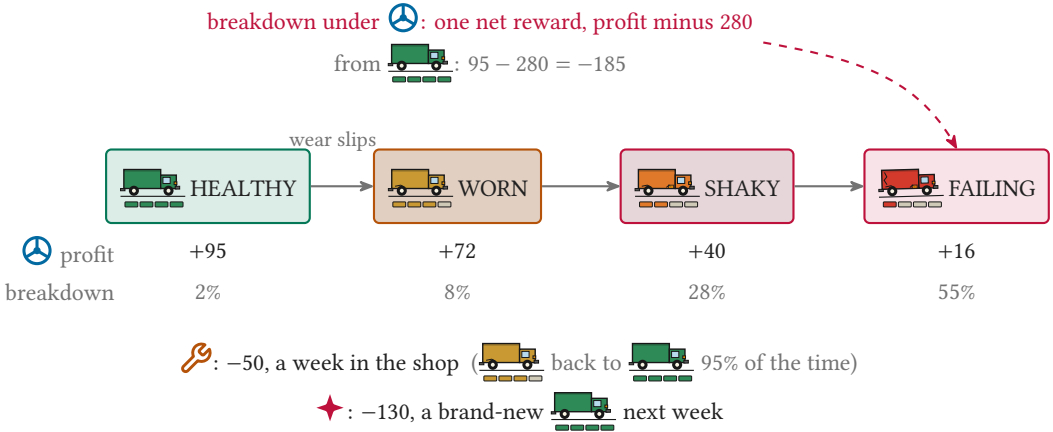
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The fleet sheet



Policies and trajectories

Playbook in, tape out

a policy (the playbook): $\pi : S \to A$, one call per state

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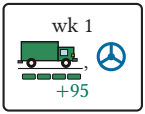
each week: $a_i = \pi(s_i)$, then $(s_{i+1}, r_i) = T(s_i, a_i)$

Playbook in, tape out

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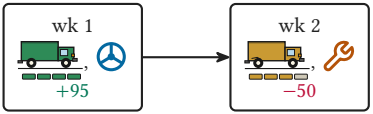


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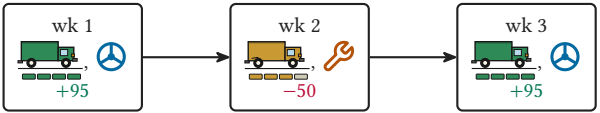


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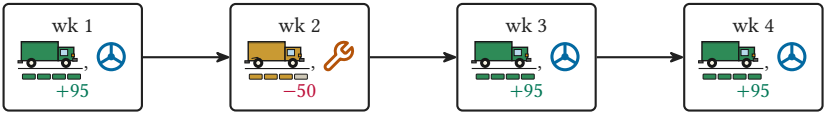


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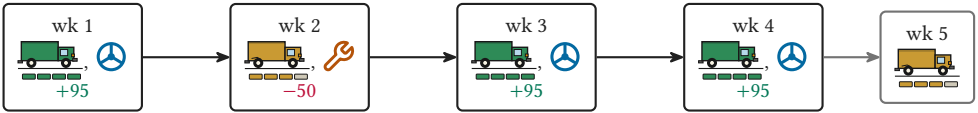


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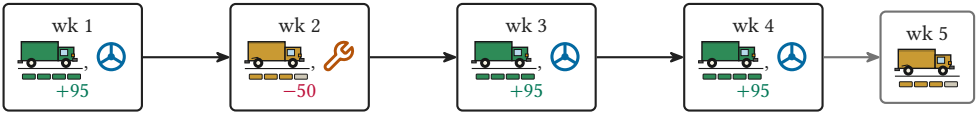
the trajectory: $s_1, a_1, r_1,\ s_2, a_2, r_2,\ \ldots$ one week per step, forever

Playbook in, tape out

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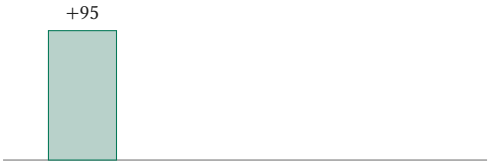
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the tape is random; another run, same start: $(\ ,\ ,\ -185) \to$ 

The goal: accumulating reward

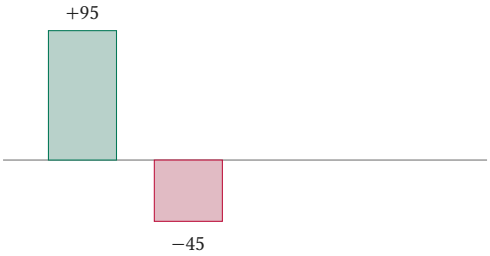
The return: one number per tape

$$G_1 = r_1$$



The return: one number per tape

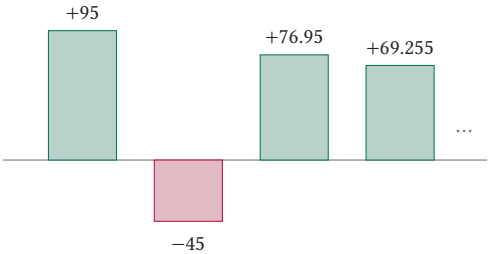
$$G_1 = \underbrace{r_1}_{\text{in full}} + \underbrace{\gamma r_2}_{\text{at 90\%}}$$



each week shrunk by $\gamma = 0.9$

The return: one number per tape

$$G_1 = \underbrace{r_1}_{\text{in full}} + \underbrace{\gamma r_2}_{\text{at 90\%}} + \underbrace{\gamma^2 r_3}_{\text{at 81\%}} + \cdots$$

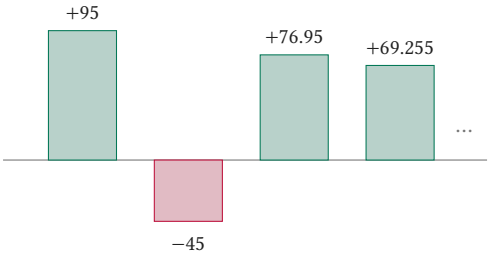


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$$G_1 = \underbrace{r_1}_{\text{in full}} + \underbrace{\gamma r_2}_{\text{at 90\%}} + \underbrace{\gamma^2 r_3}_{\text{at 81\%}} + \cdots$$

$$G_1 = 95 + 0.9\,(-50) + 0.81 \cdot 95 + 0.729 \cdot 95 + \cdots = 95 - 45 + 76.95 + 69.255 + \cdots$$

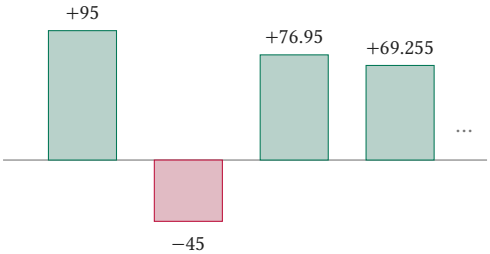


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each week shrunk
by $\gamma = 0.9$

the shrinking factors keep
the never-ending sum finite

The recurrence

$$G_i = r_i + \gamma r_{i+1} + \gamma^2 r_{i+2} + \cdots$$



The recurrence

$$G_i = r_i + \gamma\left(r_{i+1} + \gamma r_{i+2} + \cdots\right)$$



G_{i+1}, discounted once by γ

The recurrence

$$G_i = \underbrace{r_i}_{\text{this week}} + \gamma \underbrace{G_{i+1}}_{\text{everything after}}$$



G_{i+1}, discounted once by γ

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$$G_i = \underbrace{r_i}_{\text{this week}} + \gamma \underbrace{G_{i+1}}_{\text{everything after}}$$



G_{i+1}, discounted once by γ

this week's reward + $\gamma\times$ all the rest

The return is random

same policy π, same start  : run the weeks again and again



The return is random

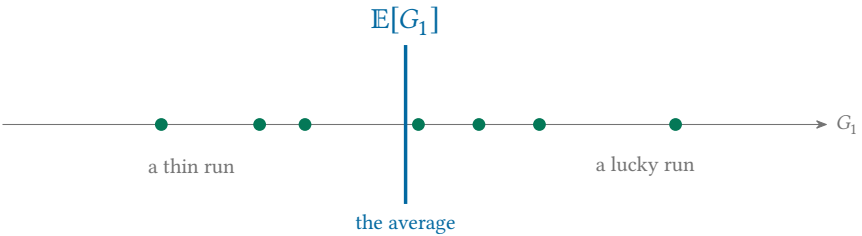
same policy π, same start  : run the weeks again and again



which G_1 is the policy's quality, if it changes every run?

The return is random

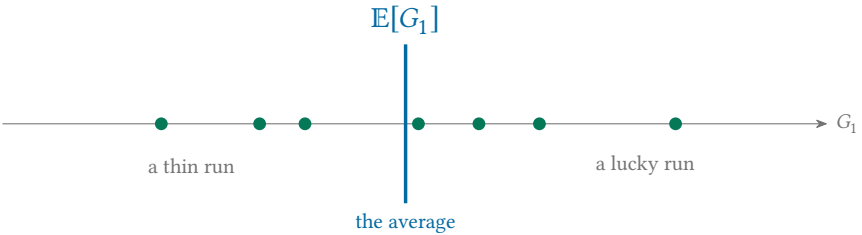
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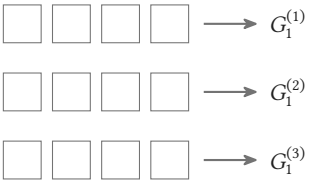
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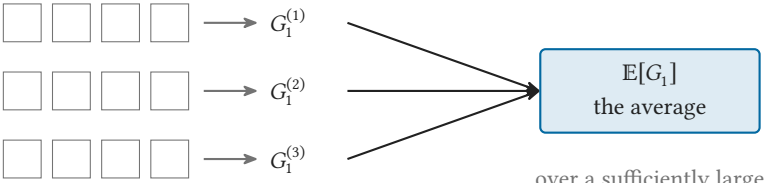
average G_1 over many runs: one honest number, $\mathbb{E}[G_1]$

Maximize the average



each run gives its own G_1

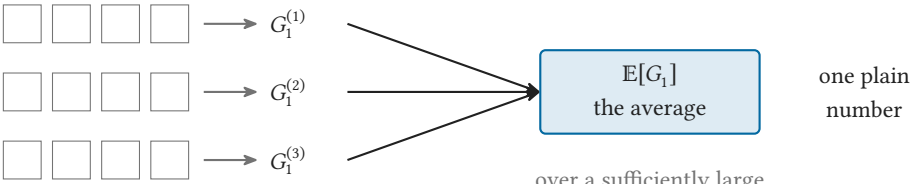
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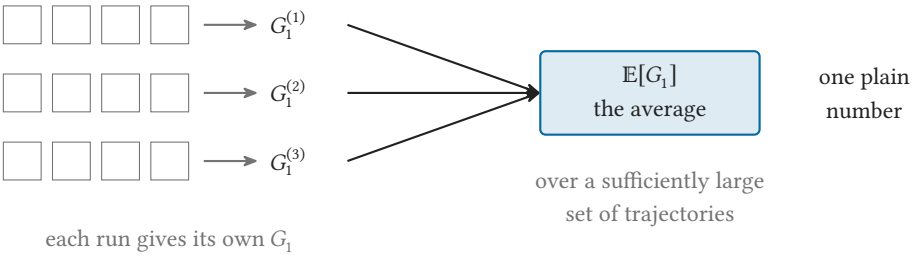
each run gives its own G_1

over a sufficiently large set of trajectories

Maximize the average



Maximize the average



the goal: find the policy π that maximizes $\mathbb{E}[G_1]$

Exploration and exploitation



MDP maze

bandit arcade

Bellman's house

SARSA maze

darts in the dark

you are here

Which call do we try while still learning?

mostly the best one so far, but explore now and then, or you never learn what you skipped

Explore or exploit?

to pile up reward you must keep choosing a call, week after week, but which?

Explore or exploit?

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EXPLOIT — take the call you currently believe is best

Explore or exploit?

to pile up reward you must keep choosing a call, week after week, but which?

EXPLOIT	take the call you currently believe is best
EXPLORE	try another, to learn what it is worth

Explore or exploit?

to pile up reward you must keep choosing a call, week after week, but which?

EXPLOIT — take the call you currently believe is best

EXPLORE — try another, to learn what it is worth

the tension: you cannot pile up reward on what you have never measured

Explore or exploit?

to pile up reward you must keep choosing a call, week after week, but which?

EXPLOIT	take the call you currently believe is best
EXPLORE	try another, to learn what it is worth

the tension: you cannot pile up reward on what you have never measured

always ☮ the van (best now), or now and then 🔧 or ✦ just to learn

The ε-greedy rule

one simple knob, here set to $\varepsilon = 0.15$

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most weeks

a share $1 - \varepsilon = 0.85$: take the best-known call (exploit)

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most weeks

a share $1 - \varepsilon = 0.85$: take the best-known call (exploit)

now and then

a small chance $\varepsilon = 0.15$: take a random call instead (explore)

The ε-greedy rule

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most weeks

a share $1 - \varepsilon = 0.85$: take the best-known call (exploit)

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a call never tried is never learned: you cannot learn the value of servicing if you never service

The ε-greedy rule

one simple knob, here set to $\varepsilon = 0.15$

most weeks

a share $1 - \varepsilon = 0.85$: take the best-known call (exploit)

now and then

a small chance $\varepsilon = 0.15$: take a random call instead (explore)

a call never tried is never learned: you cannot learn the value of servicing if you never service

ε is the dial that balances explore against exploit

The Q-function and the Bellman equation



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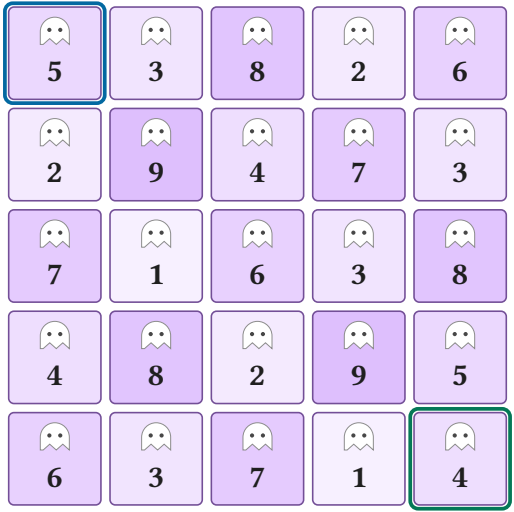
darts in the dark

you are here

Which call wins in the long run?

the one with the highest cumulative reward; the Bellman equation pins down its value $Q^*(s, a)$

START



EXIT

The scorecard $Q^\star$

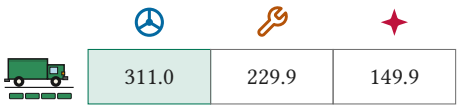
$$Q^\star(s,a) = \mathbb{E}\Big[\, G_i \;\Big|\; s_i = s,\ a_i = a \,\Big]$$

The scorecard $Q^\star$

$$Q^\star(s,a) = \mathbb{E}\Big[\, G_i \;\Big|\; \underbrace{s_i = s,\ a_i = a}_{\text{in } s\text{, play } a\text{, optimal afterwards}} \Big]$$

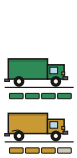
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311.0	229.9	149.9
203.4	226.1	149.9

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	311.0	229.9	149.9
	203.4	226.1	149.9
	96.5	126.4	149.9

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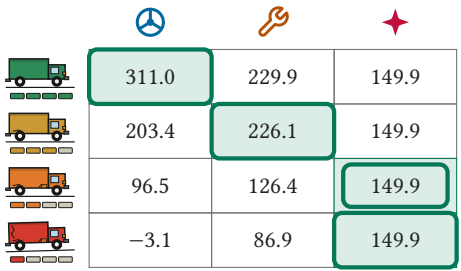
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	311.0	229.9	149.9
	203.4	226.1	149.9
	96.5	126.4	149.9
	−3.1	86.9	149.9

The scorecard $Q^\star$

$$Q^\star(s,a) = \mathbb{E}\big[\, G_i \mid \underbrace{s_i = s,\ a_i = a}_{\text{in } s\text{, play } a\text{, optimal afterwards}} \,\big]$$

$\pi^\star$: read each row, keep the call with the largest value



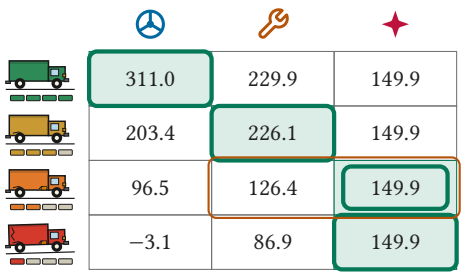
	311.0	229.9	149.9
	203.4	226.1	149.9
	96.5	126.4	149.9
	−3.1	86.9	149.9

row by row: the kept call

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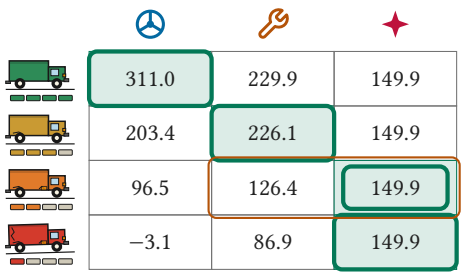
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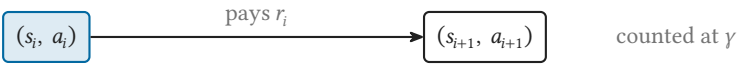
row by row: the kept call

149.9 beats 126.4 by 23.5

the optimal playbook replaces a van that still starts every morning

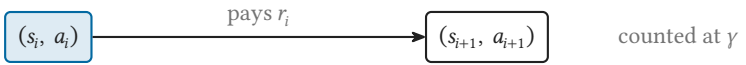
The Bellman equation

$$G_i = r_i + \gamma G_{i+1} \qquad \text{(every single week)}$$



The Bellman equation

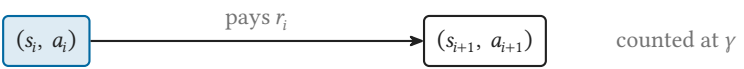
$$Q^\star(s_i, a_i) = \mathbb{E}\big[\, r_i + \gamma G_{i+1} \,\big] \qquad \text{(average both sides)}$$



The Bellman equation

$$Q^\star(s_i, a_i) = \mathbb{E}\Big[\underbrace{r_i}_{\text{this week's cash}} + \gamma \underbrace{Q^\star(s_{i+1}, a_{i+1})}_{\text{next week's value}}\Big]$$

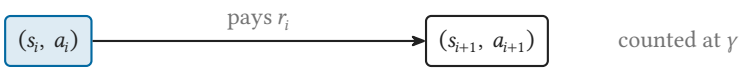
replacing G_{i+1} by its average $Q^\star(s_{i+1}, a_{i+1})$: averaging in two rounds



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one equation per cell: twelve equations pin down the table

Learning a number in the dark: Robbins-Monro



MDP maze

bandit arcade

Bellman's house

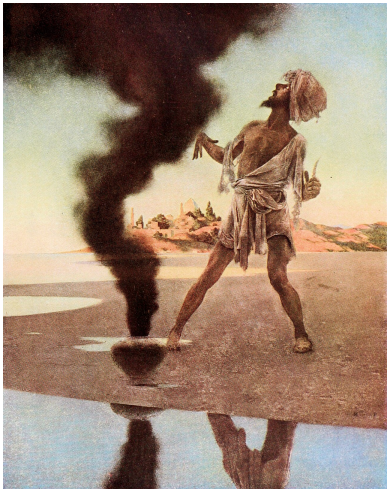
SARSA maze

darts in the dark

you are here

Why does nudging ever settle down?

it is a running average of noisy evidence; Robbins and Monro proved small shrinking steps find the truth



Maxfield Parrish, *The Arabian Nights* (1909)

Lost in the desert, you rub a lamp. The genie will set you free, on one condition: solve

$$Q = 1 + \frac{1}{Q}$$

No computer, no formula, only arithmetic.
How do you even begin?

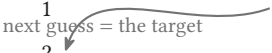
Guess a number, then let the equation correct your guess.

Guess, check, correct

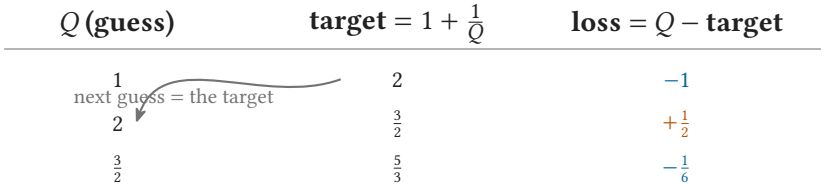
Q (guess)	target $= 1 + \frac{1}{Q}$	loss $= Q -$ target
1	2	-1

Guess, check, correct

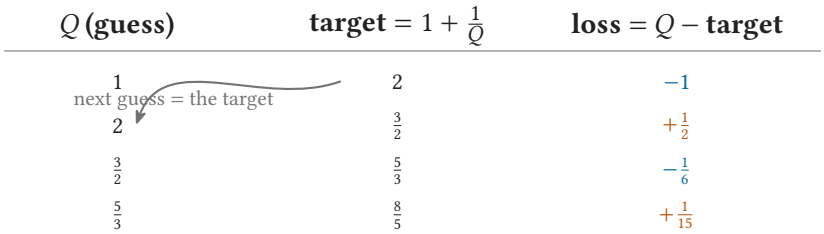
Q (guess)	**target** $= 1 + \frac{1}{Q}$	**loss** $= Q -$ **target**
1	2	-1
2	$\frac{3}{2}$	$+\frac{1}{2}$



Guess, check, correct

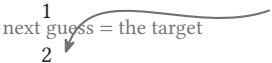


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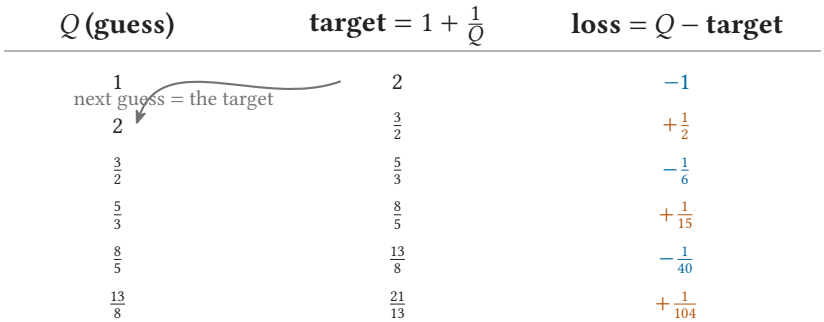


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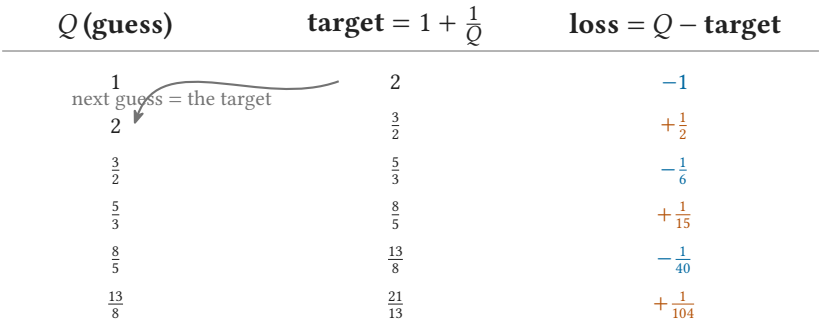
Q (guess)	target $= 1 + \frac{1}{Q}$	loss $= Q -$ target
1	2	-1
2	$\frac{3}{2}$	$+\frac{1}{2}$
$\frac{3}{2}$	$\frac{5}{3}$	$-\frac{1}{6}$
$\frac{5}{3}$	$\frac{8}{5}$	$+\frac{1}{15}$
$\frac{8}{5}$	$\frac{13}{8}$	$-\frac{1}{40}$



Guess, check, correct



Guess, check, correct



the loss shrinks toward 0, so $Q \to 1.618\ldots = \varphi$. guess equals target: solved.

The same idea, written as an algorithm

```
Q := 1                                  any first guess
repeat:
target := 1 + 1/Q                       what the equation demands
loss   := Q - target                    how far off, with sign
Q      := Q - loss                      step onto the target
until loss is (almost) 0
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Q := Q - α·loss
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make the target a *noisy* sample and let the step α shrink: that is Robbins-Monro

The Robbins-Monro iteration

to value a call we must average noisy weekly evidence, but the average is never handed to us; it arrives one week at a time. Robbins and Monro (1951) gave the recipe, and the game “darts in the dark” is its picture.

find the number $x^\star$ that equals the average of noisy samples, one at a time

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$$x \leftarrow x + \alpha_n \left(\text{sample} - x\right)$$

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this is exactly the nudge SARSA will use next, cell by cell

SARSA: learning from the logbook



MDP maze

bandit arcade

Bellman's house

SARSA maze

darts in the dark

you are here

What if nobody hands us the odds?

read the logbook: after each lived week, nudge one cell toward what it actually paid

No map, only the walking



drawing by L. Oroz, 1925

Caminante, no hay camino,
se hace camino al andar.

(trans.) Wanderer, there is no road; the road is made by walking.
Antonio Machado, Proverbios y cantares (1912)

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nobody prints the odds, so there is no map;
a good policy is a road that exists only once we walk it.

Three bookkeepers, one bad week

the odds are still hidden, so fill the () card from lived weeks

() pays **+72**

Overwriter
card = last week
72

Filing cabinet
true average of every week
72

Nudger
move 15% toward the week, a footstep correcting the road
72

Three bookkeepers, one bad week

the odds are still hidden, so fill the () card from lived weeks

() pays	+72	+72
Overwriter card = last week	72	72
Filing cabinet true average of every week	72	72
Nudger move 15% toward the week, a footstep correcting the road	72	72

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the odds are still hidden, so fill the () card from lived weeks

() pays	+72	+72	+72
Overwriter card = last week	72	72	72
Filing cabinet true average of every week	72	72	72
Nudger move 15% toward the week, a footstep correcting the road	72	72	72

Three bookkeepers, one bad week

the odds are still hidden, so fill the () card from lived weeks

() pays	+72	+72	+72	−208
Overwriter card = last week	72	72	72	−208
Filing cabinet true average of every week	72	72	72	2
Nudger move 15% toward the week, a footstep correcting the road	72	72	72	30

Three bookkeepers, one bad week

the odds are still hidden, so fill the () card from lived weeks

() pays	+72	+72	+72	−208	+72
Overwriter card = last week	72	72	72	−208	72
Filing cabinet true average of every week	72	72	72	2	16
Nudger move 15% toward the week, a footstep correcting the road	72	72	72	30	36.3

Three bookkeepers, one bad week

the odds are still hidden, so fill the () card from lived weeks

() pays	+72	+72	+72	−208	+72	
Overwriter card = last week	72	72	72	−208	72	hysterical
Filing cabinet true average of every week	72	72	72	2	16	right, but stores every card
Nudger move 15% toward the week, a footstep correcting the road	72	72	72	30	36.3	calm, and no cabinet

Three bookkeepers, one bad week

the odds are still hidden, so fill the () card from lived weeks

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Overwriter card = last week	72	72	72	−208	72	hysterical
Filing cabinet true average of every week	72	72	72	2	16	right, but stores every card
Nudger move 15% toward the week, a footstep correcting the road	72	72	72	30	36.3	calm, and no cabinet

$$q(s_i, a_i) \leftarrow q(s_i, a_i) + \alpha\big[\, r_i - q(s_i, a_i) \,\big]$$ (the Nudger, written down)

Three bookkeepers, one bad week

the odds are still hidden, so fill the () card from lived weeks

() pays	+72	+72	+72	−208	+72	
Overwriter card = last week	72	72	72	−208	72	hysterical
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$$q(s_i, a_i) \leftarrow \underbrace{q(s_i, a_i)}_{\text{the card}} + \underbrace{\underset{\sim}{\alpha}}_{0.15} \underbrace{\left[r_i - q(s_i, a_i) \right]}_{\text{the surprise}}$$

Three bookkeepers, one bad week

the odds are still hidden, so fill the () card from lived weeks

() pays	+72	+72	+72	−208	+72	
Overwriter card = last week	72	72	72	−208	72	hysterical
Filing cabinet true average of every week	72	72	72	2	16	right, but stores every card
Nudger move 15% toward the week, a footstep correcting the road	72	72	72	30	36.3	calm, and no cabinet

$$q(s_i, a_i) \leftarrow \underbrace{q(s_i, a_i)}_{\text{the card}} + \underbrace{\underset{\sim}{\alpha}}_{0.15} \underbrace{\left[r_i - q(s_i, a_i) \right]}_{\text{the surprise}}$$

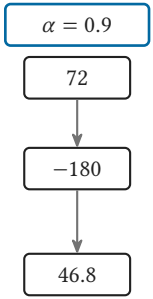
nudge, never overwrite: one bad week must not erase a year

The alpha knob

same five weeks, three step sizes

The alpha knob

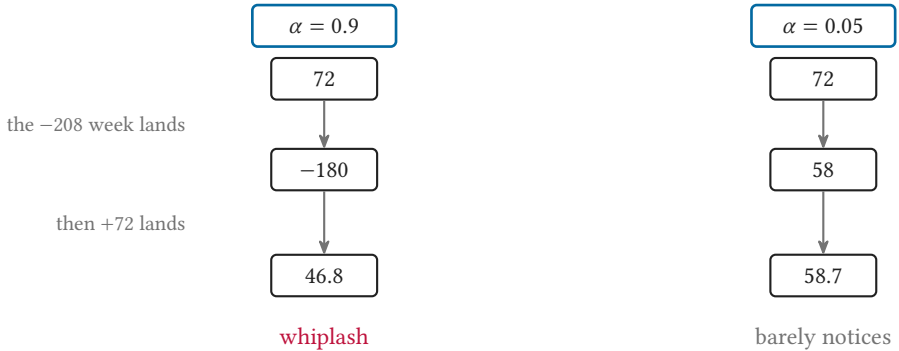
same five weeks, three step sizes



whiplash

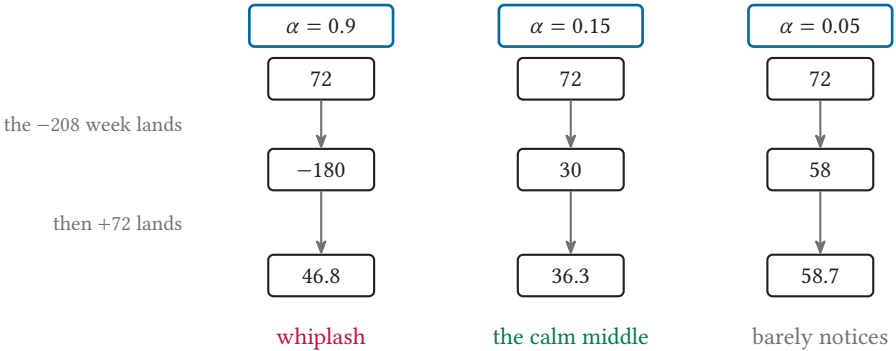
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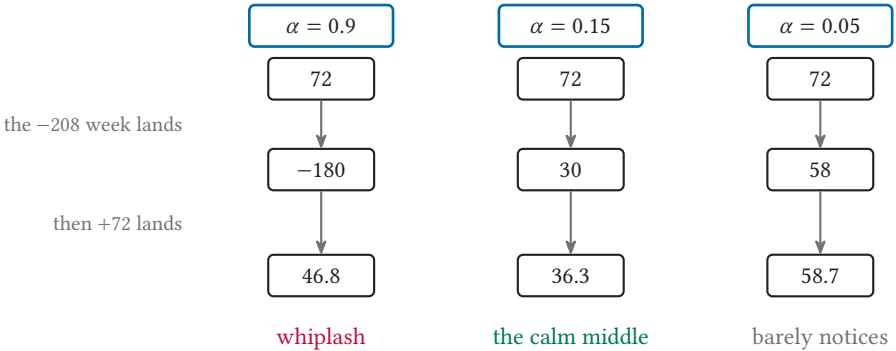
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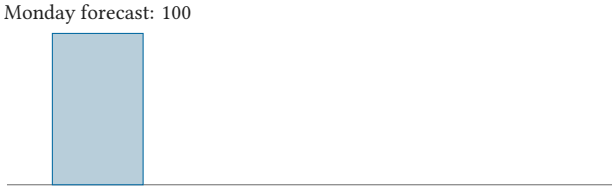
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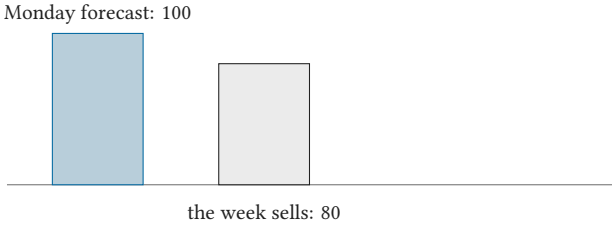


small steady steps, never overwrite

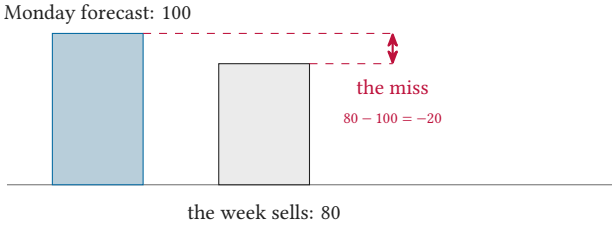
The planner's Monday



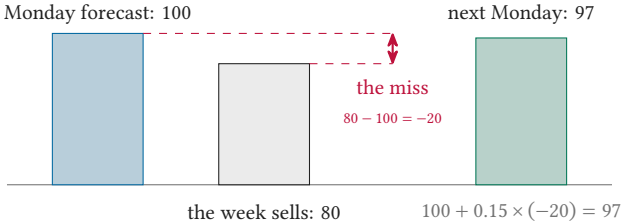
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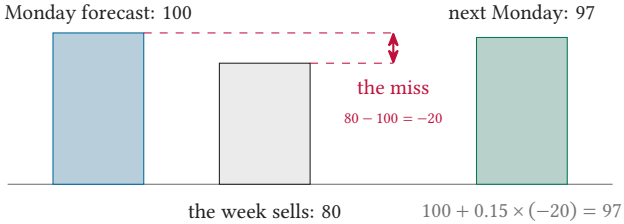
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The planner's Monday

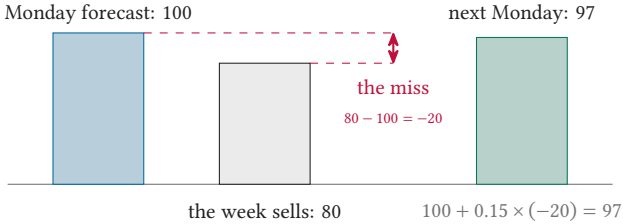


The planner's Monday



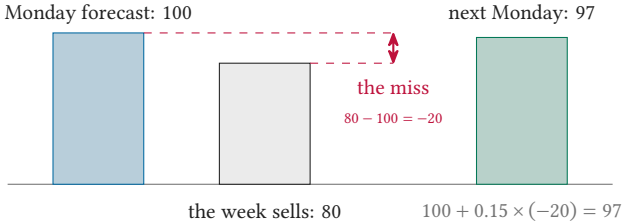
$$\text{new forecast} = \text{old forecast} + 0.15 \times \text{miss}$$

The planner's Monday



$$\text{new forecast} = \underbrace{\text{old forecast}}_{q(s_i, a_i)} + \underbrace{0.15}_{\alpha} \times \underbrace{\text{miss}}_{r_i - q(s_i, a_i)}$$

The planner's Monday

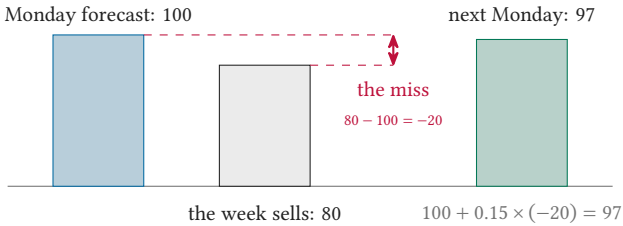


the planner's name:

exponential smoothing

$$\text{new forecast} = \underbrace{\text{old forecast}}_{q(s_i, a_i)} + \underbrace{0.15}_{\alpha} \times \underbrace{\text{miss}}_{r_i - q(s_i, a_i)}$$

The planner's Monday



the planner's name:

exponential smoothing

$$\text{new forecast} = \underbrace{\text{old forecast}}_{q(s_i, a_i)} + \underbrace{0.15}_{\alpha} \times \underbrace{\text{miss}}_{r_i - q(s_i, a_i)}$$

SARSA forecasts the value of a call the way a planner forecasts sales

The cash-only ledger lies

the nudger judged each call by one thing: the money it pays *this week*

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by cash alone



the two *worst*
calls on the sheet

The cash-only ledger lies

the nudger judged each call by one thing: the money it pays *this week*

by cash alone

: pay 130

: pay 50

the two *worst* calls on the sheet

vet $O^\star$ crowned them

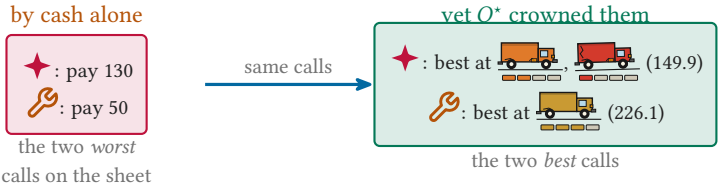
: best at , (149.9)

: best at (226.1)

the two *best* calls

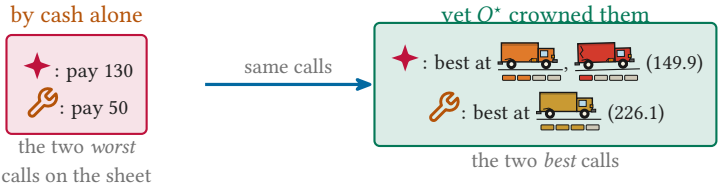
The cash-only ledger lies

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The cash-only ledger lies

the nudger judged each call by one thing: the money it pays *this week*



the worst-paying calls are the best calls, so cash cannot be the whole story

Why? A call also leaves you a van

a call does *two* things: it pays cash *now*, and it leaves you a *van* for next week

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earn +95 now , but the van ends a week more worn, a thinner future (and a breakdown leaves )

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loses cash, yet the  it leaves is worth far more than 130 down the road: that is why it is crowned

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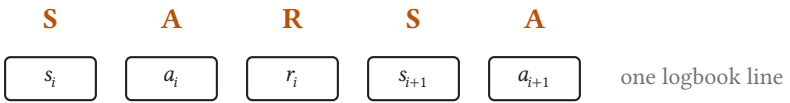


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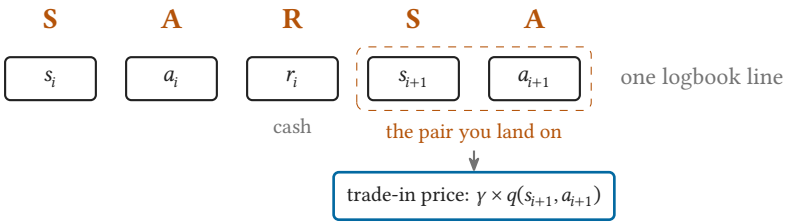
loses cash, yet the  it leaves is worth far more than 130 down the road: that is why it is crowned

a call's worth: its cash, plus the van it leaves you

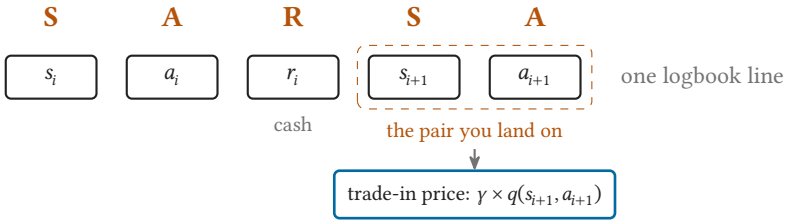
Grade the week like a trade-in



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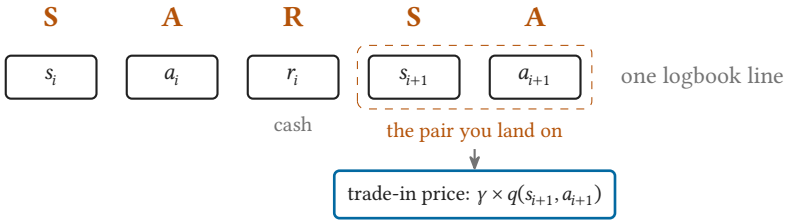


Grade the week like a trade-in



$$\text{the week's worth} = \underbrace{r_i}_{\text{cash}} + \underbrace{\gamma q(s_{i+1}, a_{i+1})}_{\text{trade-in, at your own price}}$$

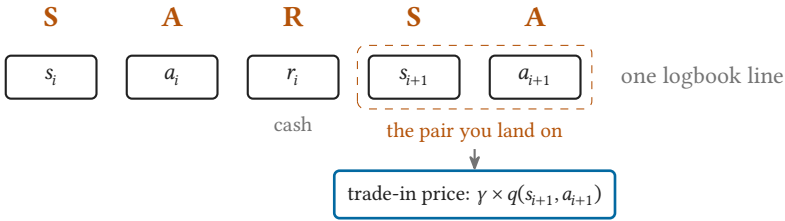
Grade the week like a trade-in



$$\text{the better target} = \underbrace{r_i + \gamma q(s_{i+1}, a_{i+1})}_{\text{cash, plus the van you keep}}$$

the surprise was your guess against cash: now aim it at this

Grade the week like a trade-in



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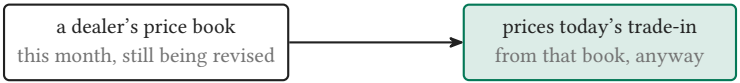
upgrade the target: from cash alone to cash + trade-in

Price it from your own book

wait: you price the leftover van with q, the very book you are still learning

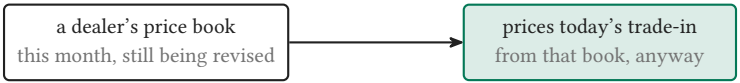
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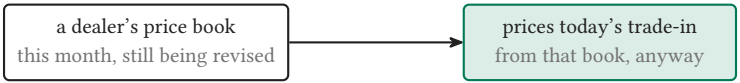
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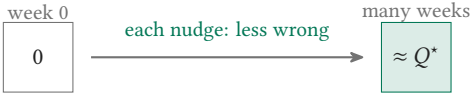
nobody refuses to price a car because the book is not final

Price it from your own book

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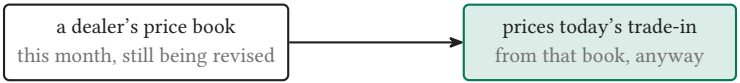


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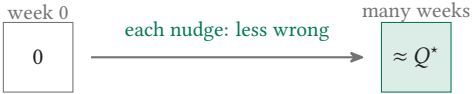


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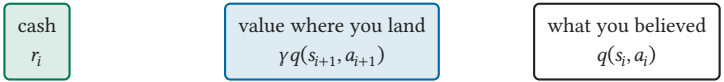


nobody refuses to price a car because the book is not final



the book is wrong today, and that is fine: every nudge makes it less wrong

Write the line yourself



Write the line yourself

cash
r_i

value where you land
$\gamma\, q(s_{i+1}, a_{i+1})$

what you believed
$q(s_i, a_i)$

$$\text{new guess} = \text{old guess} + \alpha\Big[\ ___ + \gamma\ _____ - _____\ \Big]$$

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$$q(s_i, a_i) \leftarrow q(s_i, a_i) + \alpha\Big[\, r_i + \gamma\, q(s_{i+1}, a_{i+1}) - q(s_i, a_i) \,\Big]$$

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the surprise: upgraded target minus your guess

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$\alpha = 0.15$, constant · $\varepsilon = 0.15$: a random call in 15% of weeks · a_{i+1}: the call you actually make, never a max

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$\alpha = 0.15$, constant · $\varepsilon = 0.15$: a random call in 15% of weeks · a_{i+1}: the call you actually make, never a max

you did not memorize SARSA: you wrote it

Bellman becomes SARSA

$$Q^\star(s_i, a_i) = \mathbb{E}\Big[\, r_i + \gamma Q^\star(s_{i+1}, a_{i+1}) \,\Big]$$

start from Bellman: an average over all the weeks that could happen

Bellman becomes SARSA

$$Q^\star(s_i, a_i) \approx \underbrace{r_i + \gamma Q^\star(s_{i+1}, a_{i+1})}_{\text{the one week that arrived, not the average}}$$

erase the average: keep the single week that actually arrived

Bellman becomes SARSA

$$q(s_i, a_i) \approx r_i + \gamma q(s_{i+1}, a_{i+1})$$ (your book, not $Q^\star$)

erase the unknown truth $Q^\star$: look the value up in your own book q

Bellman becomes SARSA

$$q(s_i, a_i) \leftarrow q(s_i, a_i) + \alpha\Big[\, r_i + \gamma\, q(s_{i+1}, a_{i+1}) - q(s_i, a_i) \,\Big]$$

wrap it in the nudge: step toward the target, never jump to it

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wrap it in the nudge: step toward the target, never jump to it

Bellman, with the expectation replaced by an arriving average

One week, one nudge

twelve guesses, all start at 0

	0	0	0
	0	0	0
	0	0	0
	0	0	0

One week, one nudge

week one in the logbook: ( ,  , +95,  , )

twelve guesses, all start at 0

			
	0	0	0
	0	0	0
	0	0	0
	0	0	0

One week, one nudge

week one in the logbook: ( ,  , +95,  , )

evidence $= r_i + \gamma q($  ,  $) = 95 + 0.9 \times 0 = 95$

twelve guesses, all start at 0

			
	0	0	0
	0	0	0
	0	0	0
	0	0	0

One week, one nudge

week one in the logbook: ( , +95, )

evidence $= r_i + \gamma q($  $) = 95 + 0.9 \times 0 = 95$

surprise $= 95 - 0 = 95$

nudge: $+ 0.15 \times 95$

twelve guesses, all start at 0

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	0	0	0
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nudge: $+ 0.15 \times 95$

twelve guesses, all start at 0

one cell moves, eleven stay put

	14.25	0	0
	0	0	0
	0	0	0
	0	0	0

One week, one nudge

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twelve guesses, all start at 0

one cell moves, eleven stay put

a few hundred weeks later:
the three bands emerge,
the road underfoot



14.25	0	0
0	0	0
0	0	0
0	0	0

One week, one nudge

week one in the logbook: ( , +95, )

evidence $= r_i + \gamma q($  $) = 95 + 0.9 \times 0 = 95$

surprise $= 95 - 0 = 95$

nudge: $+\,0.15 \times 95$

twelve guesses, all start at 0

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a few hundred weeks later:
the three bands emerge,
the road underfoot

	14.25	0	0
	0	0	0
	0	0	0
	0	0	0

the cells settle below $Q^\star$ and keep jittering, but each row keeps the right call

SARSA, all together

1 you are in a state (the van's wear level: );

SARSA, all together

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SARSA, all together

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2. choose a call: usually the best you know (exploit), sometimes a random one (explore): ε-greedy;
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4. nudge your estimate of that call a little toward (this week's reward + $\gamma \times$ your estimate of the next call): the Robbins-Monro nudge toward the Bellman target;

SARSA, all together

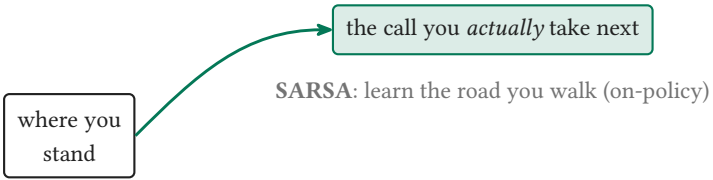
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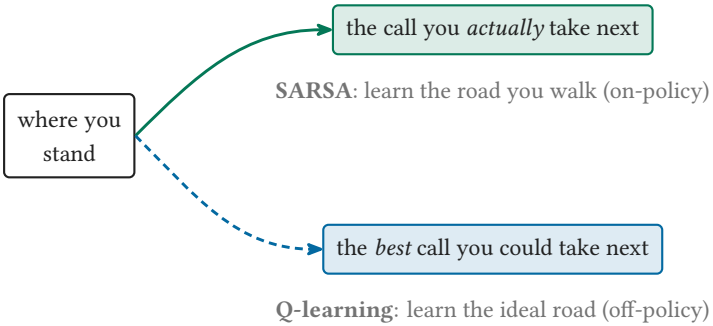
this brings together the MDP (states, calls), explore/exploit (step 2), Bellman (the target in step 4), and Robbins-Monro (the nudge in step 4)

SARSA's cousin: Q-learning

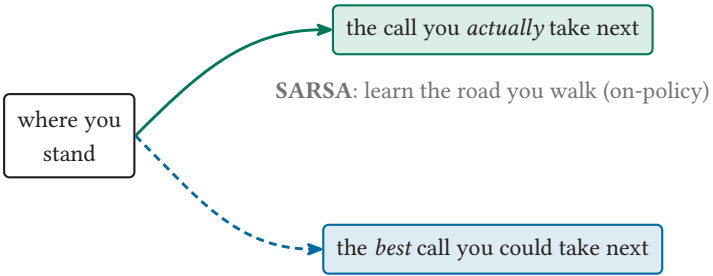


SARSA: learn the road you walk (on-policy)

SARSA's cousin: Q-learning



SARSA's cousin: Q-learning

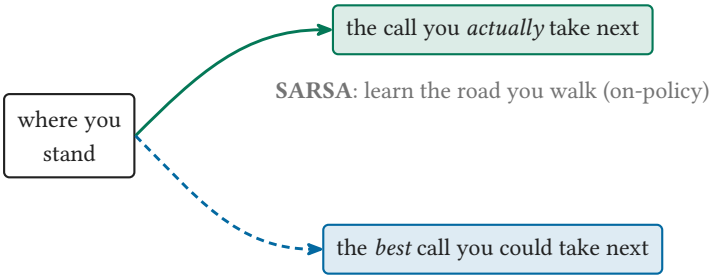


SARSA: learn the road you walk (on-policy)

Q-learning: learn the ideal road (off-policy)

Same nudge, one word changed: SARSA aims at the call it will really make next; Q-learning aims at the best call on offer, even while it wanders elsewhere.

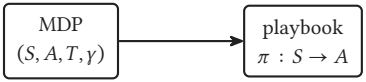
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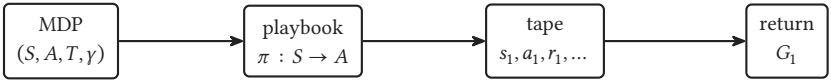
Same nudge, one word changed: SARSA aims at the call it will really make next; Q-learning aims at the best call on offer, even while it wanders elsewhere.

the road you walk, or the best road you can see

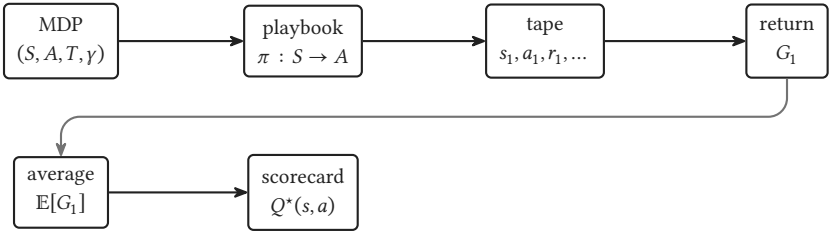
The whole story



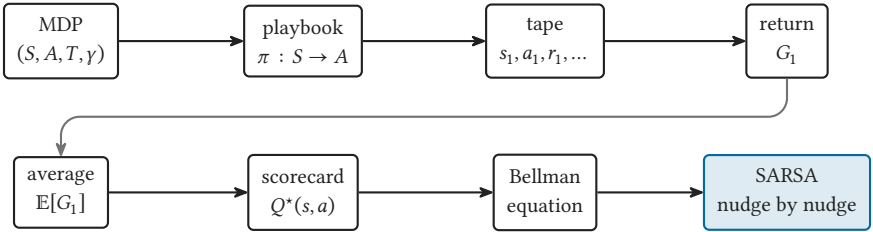
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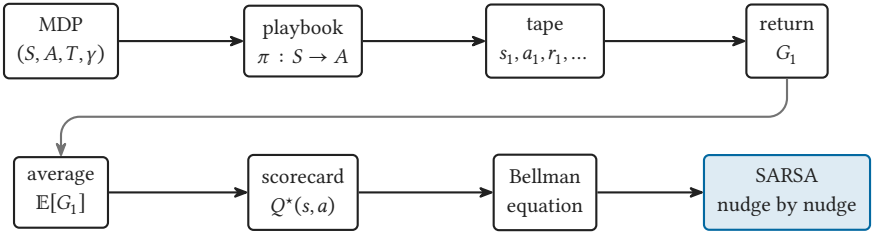
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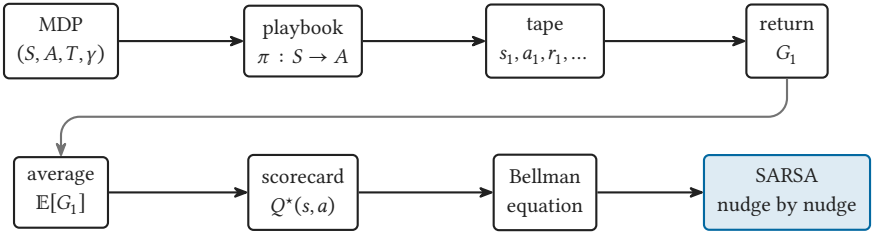


The whole story



SARSA solves the Bellman equation by averaging evidence instead of computing expectations

The whole story



SARSA solves the Bellman equation by averaging evidence instead of computing expectations

play it: `https://eth-ainit-fs26.github.io/rl-arcade/repair-or-replace/`

Appendix: why the nudge works

The running average

A_n: the average of the first n observations $x_1, \ldots, x_n$

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$$A_n = A_{n-1} + \frac{1}{n}\left(x_n - A_{n-1}\right)$$

The running average

A_n: the average of the first n observations $x_1, \ldots, x_n$

$$A_n = \underbrace{A_{n-1}}_{\text{average so far}} + \frac{1}{n} \underbrace{\left(x_n - A_{n-1}\right)}_{\text{one share of the surprise}}$$

The running average

A_n: the average of the first n observations $x_1, \ldots, x_n$

$$A_n = \underbrace{A_{n-1}}_{\text{average so far}} + \frac{1}{n} \underbrace{\left(x_n - A_{n-1}\right)}_{\text{one share of the surprise}}$$

hold α constant instead of $1/n$: each old week fades by $(1 - \alpha)$

The running average

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$$A_n = \underbrace{A_{n-1}}_{\text{average so far}} + \frac{1}{n} \underbrace{\left(x_n - A_{n-1}\right)}_{\text{one share of the surprise}}$$

hold α constant instead of $1/n$: each old week fades by $(1 - \alpha)$

SARSA is a running average of the weekly evidence